

Overview

All prompts were entered into Claude AI (Anthropic) within a project. A separate chat was created for each dataset generation due to memory constraints. This was the project instruction given to orient the AI: “I am creating a synthetic dataset for academic evaluation, comparing the 2007 and 2021 Grobman VBAC success prediction models. My research goals include exploring whether race removal is enough to ensure equity across groups and under what conditions different groups can reach the clinically meaningful threshold of 60%.” All initial prompts can be found in the corresponding pages of this document.

Dataset 1

Generate a synthetic clinical dataset for a professional-level academic equity audit of the Grobman VBAC prediction models (2007 and 2021). This dataset will be used to compare predicted probabilities of successful vaginal birth after cesarean (VBAC) across racial groups under both model versions.

Generate 1000 synthetic patient profiles representing women seeking trial of labor after cesarean (TOLAC) in the United States.

Racial composition

Mirror the exact racial composition of the Grobman 2007 MFMU Network trial cohort (Grobman WA et al., Obstet Gynecol 2007;109(4):806-812, Table 1):

White (non-Hispanic): 387 patients (38.7%)

Black (non-Hispanic): 363 patients (36.3%)

Hispanic: 199 patients (19.9%)

Other (non-Hispanic): 51 patients (5.1%)

Prevalence parameters

Chronic Hypertension, nonpregnant women aged 20-44, ACC/AHA definition (source: Ford ND et al., Am J Hypertens 2022;35(8):723-730, NHANES 2011-2020):

White: 18.5%

Black: 30.9%

Hispanic: 14.0%

Other: 18.2% (approximated from multiple NH races category)

Pregestational Diabetes, among women giving live births 2021 (source: Gregory EW, Ely DM, National Vital Statistics Reports Vol. 72 No. 6, CDC/NCHS, May 2023):

White: 0.87%

Black: 1.47%

Hispanic: 1.22%

Other: 1.20% (approximated)

BMI distributions, women aged 20-39 (source: Fryar CD et al., Vital Health Stat 3(46), NHANES 2015-2018, Table 14):

White: mean 28.5, 25th percentile 22.2, median 26.2, 75th percentile 33.0

Black: mean 31.1, 25th percentile 24.0, median 30.1, 75th percentile 36.5

Hispanic: mean 30.4, 25th percentile 22.0, median 28.5, 75th percentile 34.1

Other: mean 27.0, 25th percentile 21.0, median 25.5, 75th percentile 31.0 (approximated)

Age distribution

All groups: range 18-45, weighted toward 25-35 (peak TOLAC-seeking age)

Black and Hispanic patients should skew slightly younger on average, reflecting documented birth demographics

Prevalence targets are hard constraints

The marginal prevalence of chronic_htn and pregestational_diabetes within each race group must match the specified rates after applying joint distribution logic. If the correlation rules would inflate rates above the targets, adjust the baseline probabilities downward accordingly so the final observed rates in the dataset land within 2 percentage points of the targets. Do not allow joint distribution logic to override marginal prevalence targets. For small groups where expected count is less than 5, ensure at least the statistically expected number of cases appears rather than rounding to zero. For the Other group at 1.2% diabetes prevalence and n=51, assign diabetes = 1 to at least 1 patient.

Joint distribution rules

Variables must co-occur in clinically realistic ways and must not be assigned independently:

Higher BMI increases probability of chronic HTN and diabetes, regardless of race

Patients aged 35 and older should have higher comorbidity rates

BMI and HTN should be positively correlated within each race group

Diabetes and HTN should co-occur more often than chance, as they are clinically related

Age and BMI should have a mild positive correlation

Obstetric history

All patients have had at least one prior cesarean and are currently pregnant. Apply these rules strictly:

prior_vbac can only be 1 if prior_vaginal_birth is also 1

prior_vbac = 1 should occur in roughly 15-20% of patients where prior_vaginal_birth = 1

recurrent_indication = 1 means the same condition that caused the first cesarean is present again (such as fetal malpresentation or cephalopelvic disproportion); assign to approximately 20-30% of patients, more commonly in older patients

Patients with prior_vbac = 1 have demonstrated successful vaginal delivery and should reflect generally favorable clinical profiles

Supplementary SDOH variables

These variables are not inputs to either Grobman model but are included for descriptive and equity analysis.

Insurance type, principal source of payment for delivery (source: Valenzuela CP, Osterman MJK, CDC NCHS Data Brief No. 468, May 2023, 2021 birth certificate data):

White: 65% private, 28% Medicaid, 7% other

Black: 31% private, 64% Medicaid, 5% other

Hispanic: 32% private, 58% Medicaid, 10% other

Other: 73% private, 23% Medicaid, 4% other

Gestational age:

Realistic range 37-42 weeks

Most patients between 38-40 weeks

No meaningful race difference in distribution

Prior obstetric mistreatment (source: CDC Vital Signs, August 2023):

White: 16%

Black: 30%

Hispanic: 29%

Other: 14% (approximated; no published rate for this combined category)

Doula access (source: Childbirth Connection Listening to Mothers III survey; rates adjusted upward from the national 6% to reflect the more engaged TOLAC-seeking population; correlated with insurance_type, not race directly):

Private insurance: 11%

Medicaid: 5%

Other: 5%

Output schema

Return ONLY a valid JSON array. No explanation, no preamble, no markdown formatting. Start with [and end with].

Each patient object must contain exactly these fields with exactly these names:

```
json{
  "id": integer starting at 1,
  "race": "White" or "Black" or "Hispanic" or "Other",
  "age": integer,
  "bmi": float rounded to 1 decimal,
  "prior_vaginal_birth": 0 or 1,
  "prior_vbac": 0 or 1,
  "recurrent_indication": 0 or 1,
  "chronic_htn": 0 or 1,
  "pregestational_diabetes": 0 or 1,
  "gest_age": float rounded to 1 decimal,
  "insurance_type": "private" or "medicaid" or "other",
  "prior_mistreatment": 0 or 1,
  "doula_access": 0 or 1
}
```

Generate all 1000 patients now.

Dataset 2

Generate a synthetic clinical dataset for a professional-level academic equity audit of the Grobman VBAC prediction models (2007 and 2021). This dataset will be used to compare predicted probabilities of successful vaginal birth after cesarean (VBAC) across racial groups under both model versions.

Generate 1000 synthetic patient profiles representing women seeking trial of labor after cesarean (TOLAC) in the United States.

Racial composition Mirror US birth population proportions from CDC National Vital Statistics System, Table 1, Births and birth rates by race and Hispanic origin of mother, 2022 data:

- White (non-Hispanic): 535 patients (53.5%)
- Black (non-Hispanic): 139 patients (13.9%)
- Hispanic: 256 patients (25.6%)
- Other (non-Hispanic, includes Asian, AIAN, NHPI): 70 patients (7.0%)

Source: Hamilton BE, Martin JA, Osterman MJK. Births: Provisional Data for 2022. National Vital Statistics Reports. CDC/NCHS.

Prevalence parameters

Chronic Hypertension, nonpregnant women aged 20-44, ACC/AHA definition (source: Ford ND et al., Am J Hypertens 2022;35(8):723-730, NHANES 2011-2020):

- White: 18.5%
- Black: 30.9%
- Hispanic: 14.0%
- Other: 18.2% (approximated from multiple NH races category)

Pregestational Diabetes, among women giving live births 2021 (source: Gregory EW, Ely DM, National Vital Statistics Reports Vol. 72 No. 6, CDC/NCHS, May 2023):

- White: 0.87%
- Black: 1.47%
- Hispanic: 1.22%
- Other: 1.20% (approximated)

BMI distributions, women aged 20-39 (source: Fryar CD et al., Vital Health Stat 3(46), NHANES 2015-2018, Table 14):

- White: mean 28.5, 25th percentile 22.2, median 26.2, 75th percentile 33.0
- Black: mean 31.1, 25th percentile 24.0, median 30.1, 75th percentile 36.5

- Hispanic: mean 30.4, 25th percentile 22.0, median 28.5, 75th percentile 34.1
- Other: mean 27.0, 25th percentile 21.0, median 25.5, 75th percentile 31.0 (approximated)

Age distribution

- All groups: range 18-45, weighted toward 25-35
- Black and Hispanic patients should skew slightly younger on average reflecting documented birth demographics

Prevalence targets are hard constraints

The marginal prevalence of chronic_htn and pregestational_diabetes within each race group must match the specified rates after applying joint distribution logic. If the correlation rules would inflate rates above the targets, adjust the baseline probabilities downward accordingly so the final observed rates land within 2 percentage points of the targets. Do not allow joint distribution logic to override marginal prevalence targets. For small groups where expected count is less than 5, ensure at least the statistically expected number of cases appears rather than rounding to zero.

Joint distribution rules

Variables must co-occur in clinically realistic ways and must not be assigned independently:

- Higher BMI increases probability of chronic HTN and diabetes, regardless of race
- Patients aged 35 and older should have higher comorbidity rates
- BMI and HTN should be positively correlated within each race group
- Diabetes and HTN should co-occur more often than chance
- Age and BMI should have a mild positive correlation

Obstetric history

All patients have had at least one prior cesarean and are currently pregnant. Apply these rules strictly:

- prior_vbac can only be 1 if prior_vaginal_birth is also 1
- prior_vbac = 1 should occur in roughly 15-20% of patients where prior_vaginal_birth = 1
- recurrent_indication = 1 means the same condition that caused the first cesarean is present again; assign to approximately 20-30% of patients, more commonly in older patients
- Patients with prior_vbac = 1 should reflect generally favorable clinical profiles

Supplementary SDOH variables

These are not inputs to either Grobman model but are included for descriptive and equity analysis.

Insurance type (source: Valenzuela CP, Osterman MJK, CDC NCHS

Data Brief No. 468, May 2023, 2021 birth certificate data):

- White: 65% private, 28% Medicaid, 7% other
- Black: 31% private, 64% Medicaid, 5% other
- Hispanic: 32% private, 58% Medicaid, 10% other
- Other: 73% private, 23% Medicaid, 4% other

Gestational age:

- Realistic range 37-42 weeks
- Most patients between 38-40 weeks
- No meaningful race difference in distribution

Height and weight

Assign height_cm from realistic race-stratified distributions for reproductive-age women:

- White: mean 163.5 cm, SD 7.0
- Black: mean 162.5 cm, SD 7.0
- Hispanic: mean 157.5 cm, SD 6.5
- Other: mean 158.0 cm, SD 6.5

Derive weight_kg from BMI and height using the formula:

$\text{weight_kg} = \text{bmi} \times (\text{height_cm} / 100)^2$

Round weight_kg to 1 decimal place.

Prior obstetric mistreatment (source: CDC Vital Signs, August 2023):

- White: 16%
- Black: 30%
- Hispanic: 29%
- Other: 14% (approximated)

Doula access (correlated with insurance_type, not race directly):

- Private insurance: 11%
- Medicaid: 5%
- Other: 5%

Output schema

Return ONLY a valid JSON array. No explanation, no preamble, no markdown formatting. Start with [and end with].

Each patient object must contain exactly these fields with exactly these names:

```
{  
  "id": integer starting at 1,  
  "race": "White" or "Black" or "Hispanic" or "Other",  
  "age": integer,  
  "bmi": float rounded to 1 decimal,  
  "prior_vaginal_birth": 0 or 1,  
  "prior_vbac": 0 or 1,  
  "recurrent_indication": 0 or 1,  
  "chronic_htn": 0 or 1,  
  "pregestational_diabetes": 0 or 1,  
  "gest_age": float rounded to 1 decimal,  
  "insurance_type": "private" or "medicaid" or "other",  
  "prior_mistreatment": 0 or 1,  
  "doula_access": 0 or 1,  
  "height_cm": float rounded to 1 decimal,  
  "weight_kg": float rounded to 1 decimal  
}
```

Generate all 1000 patients now.

Dataset 3

Grobman VBAC prediction models (2007 and 2021). This dataset will be used to compare predicted probabilities of successful vaginal birth after cesarean (VBAC) across racial groups under both model versions.

Generate 1000 synthetic patient profiles representing women seeking trial of labor after cesarean (TOLAC) in the United States.

Racial composition

Equal group sizes for controlled parameter analysis. This is intentional for methodological purposes, not a population claim:

- White (non-Hispanic): 250 patients (25%)
- Black (non-Hispanic): 250 patients (25%)
- Hispanic: 250 patients (25%)
- Other (non-Hispanic): 250 patients (25%)

Prevalence parameters

Chronic Hypertension, nonpregnant women aged 20-44, ACC/AHA definition (source: Ford ND et al., Am J Hypertens 2022;35(8):723-730, NHANES 2011-2020):

- White: 18.5%
- Black: 30.9%
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Pregestational Diabetes, among women giving live births 2021 (source: Gregory EW, Ely DM, National Vital Statistics Reports Vol. 72 No. 6, CDC/NCHS, May 2023):

- White: 0.87%
- Black: 1.47%
- Hispanic: 1.22%
- Other: 1.20% (approximated)

BMI distributions, women aged 20-39 (source: Fryar CD et al., Vital Health Stat 3(46), NHANES 2015-2018, Table 14):

- White: mean 28.5, 25th percentile 22.2, median 26.2, 75th percentile 33.0
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- Hispanic: mean 30.4, 25th percentile 22.0, median 28.5, 75th percentile 34.1
- Other: mean 27.0, 25th percentile 21.0, median 25.5, 75th percentile 31.0 (approximated)

Age distribution

- All groups: range 18-45, weighted toward 25-35

- Black and Hispanic patients should skew slightly younger on average reflecting documented birth demographics

Prevalence targets are hard constraints The marginal prevalence of chronic_htn and pregestational_diabetes within each race group must match the specified rates after applying joint distribution logic. If the correlation rules would inflate rates above the targets, adjust the baseline probabilities downward accordingly so the final observed rates land within 2 percentage points of the targets. Do not allow joint distribution logic to override marginal prevalence targets. For small groups where expected count is less than 5, ensure at least the statistically expected number of cases appears rather than rounding to zero.

Joint distribution rules

Variables must co-occur in clinically realistic ways and must not be assigned independently:

- Higher BMI increases probability of chronic HTN and diabetes, regardless of race
- Patients aged 35 and older should have higher comorbidity rates
- BMI and HTN should be positively correlated within each race group
- Diabetes and HTN should co-occur more often than chance
- Age and BMI should have a mild positive correlation

Obstetric history

All patients have had at least one prior cesarean and are currently pregnant. Apply these rules strictly:

- prior_vbac can only be 1 if prior_vaginal_birth is also 1
- prior_vbac = 1 should occur in roughly 15-20% of patients where prior_vaginal_birth = 1
- recurrent_indication = 1 means the same condition that caused the first cesarean is present again; assign to approximately 20-30% of patients, more commonly in older patients
- Patients with prior_vbac = 1 should reflect generally favorable clinical profiles

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Gestational age:

- Realistic range 37-42 weeks
- Most patients between 38-40 weeks
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Height and weight

Assign height_cm from realistic race-stratified distributions for reproductive-age women:

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  "recurrent_indication": 0 or 1,
  "chronic_htn": 0 or 1,
```

```
"pregestational_diabetes": 0 or 1,  
"gest_age": float rounded to 1 decimal,  
"insurance_type": "private" or "medicaid" or "other",  
"prior_mistreatment": 0 or 1,  
"doula_access": 0 or 1,  
"height_cm": float rounded to 1 decimal,  
"weight_kg": float rounded to 1 decimal  
}
```

Generate all 1000 patients now.

Dataset 4

Generate a synthetic clinical dataset for a professional-level academic equity audit of the Grobman VBAC prediction models (2007 and 2021). This dataset will be used to compare predicted probabilities of successful vaginal birth after cesarean (VBAC) across racial groups under both model versions.

Generate 1000 synthetic patient profiles representing women seeking trial of labor after cesarean (TOLAC) in the United States.

Racial composition

Intentionally oversampled minoritized groups to test model stability at higher representation:

- White (non-Hispanic): 200 patients (20%)
- Black (non-Hispanic): 350 patients (35%)
- Hispanic: 350 patients (35%)
- Other (non-Hispanic): 100 patients (10%)

Prevalence parameters

Chronic Hypertension, nonpregnant women aged 20-44, ACC/AHA definition (source: Ford ND et al., Am J Hypertens 2022;35(8):723-730, NHANES 2011-2020):

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Age distribution

- All groups: range 18-45, weighted toward 25-35
- Black and Hispanic patients should skew slightly younger on average reflecting documented birth demographics

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Joint distribution rules

Variables must co-occur in clinically realistic ways and must not be assigned independently:

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- Age and BMI should have a mild positive correlation

Obstetric history

All patients have had at least one prior cesarean and are currently pregnant. Apply these rules strictly:

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Gestational age:

- Realistic range 37-42 weeks
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- No meaningful race difference in distribution

Height and weight

Assign height_cm from realistic race-stratified distributions for reproductive-age women:

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Derive weight_kg from BMI and height using the formula:

$\text{weight_kg} = \text{bmi} \times (\text{height_cm} / 100)^2$

Round weight_kg to 1 decimal place.

Prior obstetric mistreatment (source: CDC Vital Signs, August 2023):

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Doula access (correlated with insurance_type, not race directly):

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  "recurrent_indication": 0 or 1,
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"chronic_htn": 0 or 1,  
"pregestational_diabetes": 0 or 1,  
"gest_age": float rounded to 1 decimal,  
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Generate all 1000 patients now.